Richard Varhelyi

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Education

University of Bonn — Bonn, Germany
Master of Science – Mobile Robotics — 2025 – 2027
Coursework: Mobile Robotics, Computer Vision, Deep Learning for Robotics, Robot Learning, Robot Mapping, Robot Operating Systems, SLAM

Óbuda University — Budapest, Hungary
Bachelor of Science – Computer Science & Engineering — 2021 – 2025
Coursework: Data Structures, Algorithms, Databases, Discrete Mathematics, Linear Algebra, Calculus, Machine Learning, Software Design and Development
Thesis: *Neural Network Based Vehicle Segmentation on Aerial Imagery*

Experience

Research Assistant | University of Bonn — Bonn, Germany
10.2026 – 09.2027

- Research Assistant at the Autonomous Intelligent Systems Group led by Prof. Sven Behnke
- Part of the NimbRo team competing in the RoboCup@Home competition
- Working on the robot's computer vision pipeline for perception and scene understanding
- Teaching Assistant for the course MA-INF 4113 - Cognitive Robotics

Fullstack Web Developer | Konasoft Kft. — Budapest, Hungary
10.2023 – 09.2025

- Worked on a law-related software used by 36 institutions and 9000 employees
- Designed and maintained databases using MSSQL, including complex queries & stored procedures
- Implemented backend logic and API endpoints using ASP.NET and Entity Framework
- Enhanced the frontend by integrating Vue.js components and API calls

Computer Vision Intern | Robert Bosch GmbH. — Budapest, Hungary
06.2024 – 12.2024

- Industry Bachelor's Thesis in cooperation with the ADAS team in Budapest
- Worked on convolutional neural networks, model training and optimization
- The aim of the research was the improvement of current techniques
- For vehicle detection and segmentation from drone captured images

Software Engineer Intern | Robert Bosch GmbH. — Budapest, Hungary
01.2023 – 07.2023

- Contributed to the Continuous Integration (CI) pipeline for embedded vehicle software (BCM)
- Worked on Volkswagen car models, ensuring software reliability and performance
- Developed Python scripts and Jenkins plugins to optimize RAM/ROM measurements

Languages

- **English:** Advanced (C1) | IELTS Academic 7.5
- **German:** Intermediate (B1)
- **Hungarian:** Native speaker